

COMPLETED AND PRELIMINARY FOUNDATIONS



EXPERIMENT-CENTERED AIMS

Measured 4D grain evolution is the scientific target

Local learning is the tool; experimental behavior defines the claims



AIM 1

Establish reliable experimental learning

Spatial reproducibility, baselines, and applicability limits



AIM 2

Establish source-aware metric comparability

Observation-rule perturbations and paired-fidelity synthesis



AIM 3

Interpret experimentally learned evolution rules

Validated responses advance toward physical attribution